MALIK JIRASEK

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EDUCATION

Oct 2025 – Present

Master of Science in Physics, University of Tübingen, Germany
Thesis: *"Keldysh-field-theory approach to the quantum Fisher information"*
Supervisor: Igor Lesanovsky

Aug 2020 – Jul 2025

Bachelor of Science in Physics, University of Tübingen, Germany
Final grade: 1.04 / 5.0 (German system; 1.0 = excellent)
Thesis: *"Quantum Sensing beyond the Heisenberg Limit with the Boundary Time Crystal"*
Supervisor: Igor Lesanovsky

RESEARCH EXPERIENCE

Nov 2025 – Present

Master of Science Thesis Research, University of Tübingen, Germany

- Developing a Keldysh-field-theory description of the quantum Fisher information in open quantum systems, with a focus on scalable sensor architectures.
- Demonstrating genuine quantum advantage in driven-dissipative many-body sensor architectures.

Aug 2025 – Present

Working Student, Bosch eBike Systems, Germany

- Implementing a Machine Learning pipeline for battery cell aging predictions, using BiLSTM networks and TensorFlow.
- Collaborating with the engineering team to integrate the model into the production workflow.

Oct 2024 – Jul 2025

Bachelor of Science Thesis Research, University of Tübingen, Germany

- Demonstrated sub-Heisenberg scaling of precision in optical-phase estimation using a boundary time crystal as a light source.
- Designed and numerically benchmarked realistic measurement protocols with QuTiP simulations.
- Results incorporated into a manuscript submitted to *Physical Review Letters* (under review).

Aug 2024 – Oct 2024

Research Intern, Humboldt University of Berlin, Germany

- Extended a Fortran-based many-body solver to include high-precision interaction potentials for ultracold-atom tweezer arrays, achieving 10^{-8} relative precision in ground state energies.

PUBLICATIONS

In preparation

Malik Jirasek, Igor Lesanovsky, and Viktoria Noel, "Keldysh-field-theory approach to the quantum Fisher information". *In preparation*.

Preprint / under review

Malik Jirasek, Igor Lesanovsky, and Albert Cabot, "The Boundary Time Crystal as a light source for quantum enhanced sensing beyond the Heisenberg Limit". *arXiv: 2511.23416 [quant-ph]*, 2025. Submitted to *Physical Review Letters* (under review). arXiv:2511.23416

PRESENTATIONS & CONFERENCES

06 Mar 2026

Contributed Talk at DPG Spring Meeting SAMOP, Mainz

TEACHING EXPERIENCE

WS 2024/25	Advanced Quantum Mechanics tutor (University of Tübingen)
SS 2024	Quantum Mechanics tutor (University of Tübingen)
WS 2023/24	Analytical Mechanics tutor (University of Tübingen)
SS 2023	Physics 2 tutor (University of Tübingen)
WS 2022/23	Physics 1 tutor (University of Tübingen)
WS 2021/22 – WS 2024/25	Physics Lab tutor (University of Tübingen)

AWARDS & SCHOLARSHIPS

2021 & 2023	Deutschlandstipendium (merit-based; awarded to ~0.5% of enrolled students)

SKILLS & METHODS

Programming	Python (TensorFlow, QuTiP), Fortran, MATLAB (Simulink)
Simulation & Modeling	Lindblad formalism, Keldysh formalism, stochastic calculus
Tools	Linux, Git, LaTeX
Languages	German (native), English (full professional proficiency), French (beginner)

SERVICE & LEADERSHIP

2026	Student representative, faculty appointment committee (ML in physics)
2022 – 2025	Treasurer & member, physics student council

REFERENCES

Available on request.